

Joshua Chen

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Education

School of Computing and Business, Business Analytics, National University of Singapore

2023 – 2027

Dean's List AY2024/2025, GPA: 4.85, Awarded NUS Merit Scholarship

Specialisation in Machine Learning and Finance | Minor in Quantitative Finance

University-Level Coursework:

- Economics Modelling for Business Analytics (A) - SQL, R, Data Visualisation
- Data Structures and Algorithms (A+) - Java, OOP
- Programming Methodology I & II (A) - Python

Professional Experience

Data Science Intern, Synapse

01/2026 – 06/2026

- Researched and evaluated AI agent architectures, browser automation workflows, and containerized execution environments, gaining hands-on experience with tool orchestration, sandboxing, Docker deployments, and AI system reliability.
- Developed AI-powered productivity applications within a modular platform, including intelligent file management, literature retrieval and summarization workflows, and AI-driven knowledge discovery tools.
- Built a healthcare-focused RAG platform using FastAPI, React, Qdrant, dense embeddings, BM25 retrieval, and reranking to transform Healthier SG clinical protocols into a citation-backed conversational assistant with confidence-aware responses.

Business Intelligence & Analysis Intern, Maribank - Monee

05/2025 – 08/2025

- Extracted and transformed large datasets from **Hadoop**, **Spark**, and **Hive** using advanced **SQL** and **Pandas**.
- Reduced manual data preparation time by 80% through workflow automation involving Hive queries, **Airflow** scheduling, and **Apps Script** enhancements.
- Enabled fraud investigation team to process cases 93% faster by delivering automated **JavaScript-based** web scraping scripts extracting critical external data.

Data Science Intern, Zig - ComfortDelgro

08/2025 – 11/2025

- Designed and implemented an end-to-end churn prediction pipeline using 200M+ rows of raw event logs to identify users with high churn propensity and enable automated targeting.
- Processed and engineered large-scale behavioural data on **BigQuery**, transforming fragmented event streams into user-level embeddings and feature matrices.
- Trained and tuned **Random Forest** and **XGBoost** models for churn prediction, using **SHAP** for explainability.
- Visualised churn probabilities and retention insights on **Power BI**.

Personal Projects

Hack4Good | Web Development Hackathon 📄, Google Developer Group - NUS

01/2025

- Organised a team of 5 to develop an e-commerce application using **JavaScript**, **React.js**, and **CSS**
- Achieving top 10 of 100+ participants in the hackathon.

Hive | Web Development, NUS

01/2025 – 04/2025

- Developed an online social networking platform using the **Vue.js** framework with **Firebase** for real-time database management.
- Designed User friendly Interface that supported over 300 connections on the platform

Movie Recommender System | Machine Learning Project, NUS

02/2025 – 04/2025

- Designed deep learning architecture in PyTorch, combining GMF and MLP to model linear and non-linear interactions.
- Developed a **hybrid LightFM model** combining collaborative and content features, using **Adagrad** with **WARP loss** for ranking and **Truncated SVD** for dimensionality reduction.

Relevant Skills

Data Analyst Skills

Data Visualization • Data Cleaning • Tableau • Pandas • NumPy • Excel • Matplotlib • Scikit-learn

Programming Languages & Frameworks

Python • Java • JavaScript • HTML • CSS • React.js • React Native • SQL • R • Node.js

Technical Skills

Object-Oriented Programming • Product Management • Product Development Lifecycle • Git • Agile & Scrum Methodology